

AAYUSHI PATEL

Software Developer | Machine Learning & Full-Stack

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PROJECTS

AI Teacher — Link & Think

FastAPI, React+TS, ChromaDB, Mistral-7B, RAG, HuggingFace

- AI study assistant — upload PDFs, docs, videos, or URLs and chat with them; streaming answers with inline source citations via a local RAG pipeline over the HuggingFace Inference API.

CerebroScan AI — Brain Tumor MRI Classification

TensorFlow, Keras, EfficientNetB0, Grad-CAM, Streamlit

- Deep-learning pipeline classifying brain MRI scans into 4 categories using a dual-model, 2-stage transfer-learning approach, with Grad-CAM explainability and an interactive Streamlit dashboard.

QA System in Finance

Mistral-7B, Q-LoRA, Chain-of-Thought, RLHF

- Fine-tuned Mistral-7B for financial QA tasks; achieved 56% accuracy, closing the gap with GPT-3.5-turbo's baseline (60%) via Q-LoRA, instruction tuning, and RLHF under limited resources.

Cross-lingual Embeddings for Low-Resource Languages

MUSE, Word Embeddings, NLP

- Built unsupervised cross-lingual word embeddings connecting Gujarati and Bengali, reducing dependence on parallel corpora for low-resource NLP.

Recommendation System for Amazon Fashion Data

Collaborative Filtering, Flask, React

- Personalized product-suggestion engine with a responsive Flask + React UI to alleviate choice overload.

Sentiment Analysis Platform

NLP, Flask, React

- Real-time sentiment classifier (positive/neutral/negative) with a user-facing web interface.

Human Activity Recognition

LSTM, Android

- LSTM-based classifier tracking human activities (walking, running), deployed in a mobile app for real-time monitoring and visualization.

SUMMARY

Software Developer at IBM and M.Tech graduate in Machine Learning (DA-IICT), with hands-on experience across the ML lifecycle — fine-tuning LLMs, building computer-vision pipelines, and shipping full-stack applications with FastAPI and React. I like turning research-leaning ML ideas into things that actually run in production.

EXPERIENCE

Software Developer at IBM

Jul 2024 – Present Gift City, Gandhinagar

- As a software developer at IBM I have contributed to feature development and resolved critical bugs on IBM Maximo to improve system functionality and user experience. Alongside core work, collaborated with the team on a side project building a RAG pipeline over Maximo's codebase, integrated with MCP to enable context-aware code assistance.

Software Developer Intern at IBM

Jan 2024 – Jul 2024 Gift City, Gandhinagar

- As a Software Developer Intern at IBM, I focused on fixing product bugs and also contributed to refactoring approximately 45 percent of the automation scripts.

ACHIEVEMENTS

 **1st Rank, Smart Gujarat Hackathon**
Medium Scale Category, organised by SSIP Cell

 **TCS CodeVita**
Ranked 2100 in Round 2

 **Google Code Jam I/O Women**
Ranked 4664

 **GATE CS**
AIR 4952 — strong theoretical grounding in computer science

STRENGTHS & SKILLS

Languages & Libraries

Python

Pandas

Numpy

Scikit-learn

Seaborn

TensorFlow

Keras

Matplotlib

SQL

React

Flask

Java

LangChain

Tools & Technologies

Jupyter Notebook

Anaconda

Android Studio

Eclipse

Git

Excel

Soft Skills

Problem-solving

Teamwork

Leadership

Adaptability

AREAS OF INTEREST

RAG

Generative AI

Natural Language Processing (NLP)

Machine Learning

Deep Learning

Large Language Models (LLMs)

Recommendation System

Data Mining

Information Retrieval

Research

LoRA

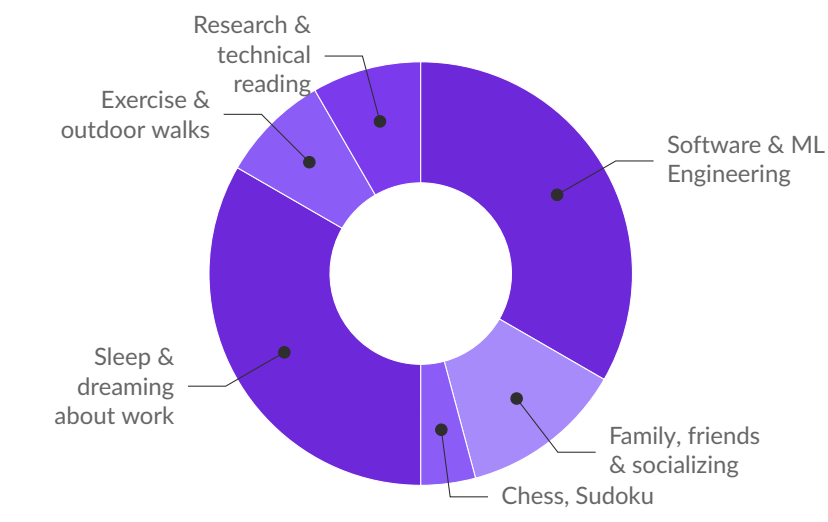
QLoRA

TECHNICAL WRITING

Author of 20+ ML-focused technical articles on Medium, breaking down LLMs, RAG, and recommender systems with code and visuals for technical and non-technical readers alike. Selected posts:

- RAG Architecture Deep Dive: The Full Pipeline Explained
- Fine-Tuning of LLM
- Instruction Tuning: Teaching LLMs to Follow Human Directions
- Session-Based & Sequential Recommenders: What You Just Did Matters Most

A DAY IN MY LIFE



EDUCATION

M.Tech, ICT (ML Specialization)
DA-ICT, Gandhinagar
2022 - 2024

B.E., Computer Engineering
LDRP-ITR
2018 - 2022

CERTIFICATIONS

- Machine Learning in Production [DeepLearning.AI]
- Generative AI with Large Language Models [DeepLearning.AI]
- Fast Prototyping of GenAI Apps with Streamlit [DeepLearning.AI]
- AI Prompting for Everyone [DeepLearning.AI]
- Fine-tuning RL for LLMs: Intro to Post-training [DeepLearning.AI]
- Retrieval Augmented Generation (RAG) [DeepLearning.AI]
- Data Science Foundations - Level 1 [IBM]
- Trustworthy AI and AI Ethics [IBM]
- Docker Essentials: A Developer Introduction [IBM]
- Prompt Engineering for AI Bootcamp 2025 [Udemy]
- Machine Learning A-Z [Udemy]

LANGUAGES

English

Gujarati

Hindi

HOBBIES

Sudoku

Chess

Reading